AFEKA 30 | Computer Science

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Multimodal Behavioral Cloning for a Robotic Arm Using Mamba and OpenCLIP

1. Problem description

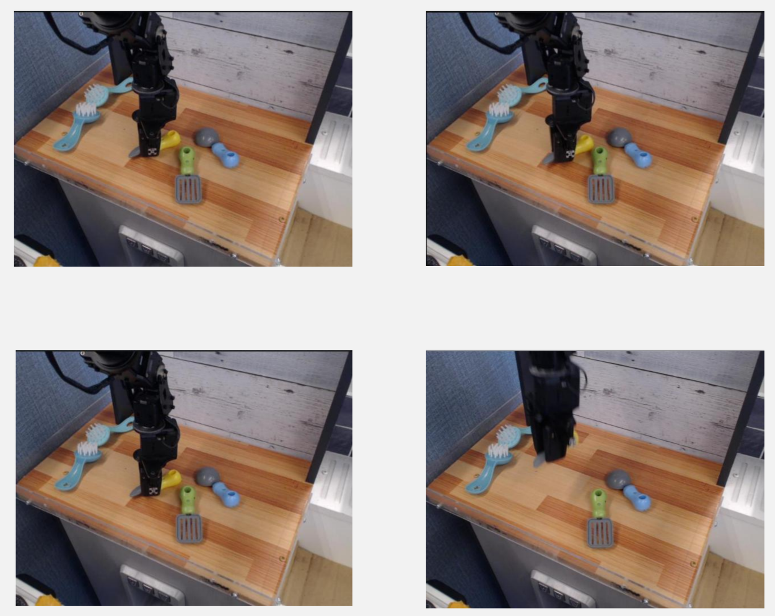
Raw Visual Complexity
Cluttered scenes with multiple candidate objects.

Silent Failures
Robot-state signals alone may miss visual failures

Temporal Dependency
Correct actions require context across consecutive frames

Cluttered scenes with multiple candidate objects

State signals alone may miss visual failures



Approach → Align →Grasp →Lift
Successful manipulation requires context across consecutive frames

2. The Solution

Behavioral Cloning
Predict the next continuous 7-DOF robot action from visual and robot-state context.

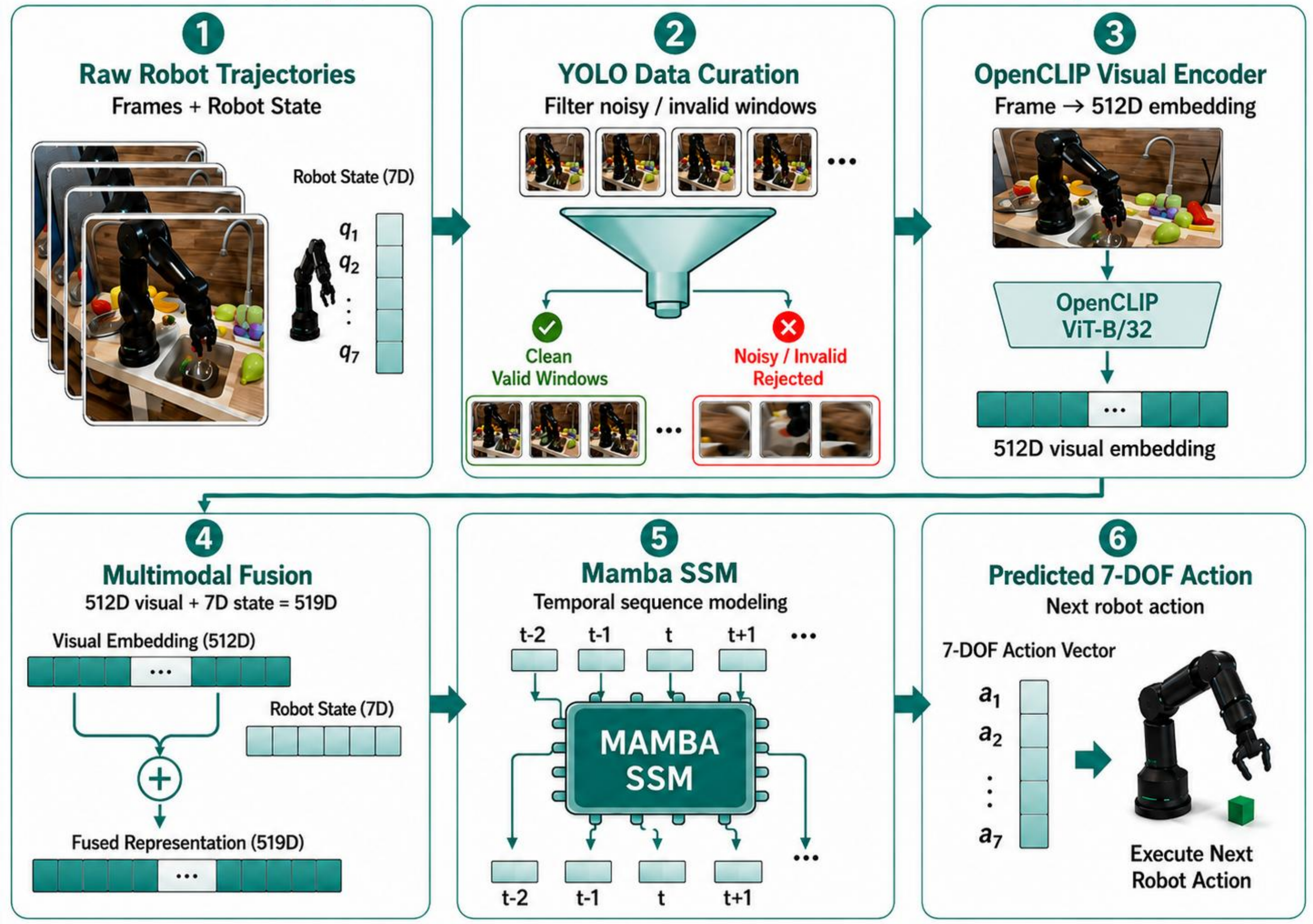
Curated Training Data
Apply YOLO-based filtering to remove invalid trajectories and retain reliable motion windows.

Multimodal Representation
Encode each frame with OpenCLIP and combine its visual embedding with the corresponding robot state.

Efficient Temporal Modeling
Use Mamba SSM to learn motion dependencies across consecutive timesteps.

3. Architecture

A 6-stage pipeline for multimodal 7-DOF robot-action prediction.



4. Engineering challenge

Reducing Silent Failures
Combine YOLO-based filtering with robot-state checks to retain reliable motion windows and reject visually invalid executions.

Multimodal Data Alignment
Synchronize each 512D visual embedding with the corresponding 7D robot-state vector to create a unified 519D representation.

Efficient Training Workflow
Precompute OpenCLIP embeddings outside the training loop to reduce repeated visual processing during Mamba training.